FOUNDATION FOR ADVANCED ROBOTICS AND AI

https://www.youtube.com/watch?v=lNuOj33pyGI

https://www.youtube.com/watch?v=KPVoBCipELk

Laws of Robotics

Isaac Asimov's three classic laws are:

- A robot may not injure a human being or, through inaction, allow a human being to come to harm.
- A robot must obey the orders given it by human beings except where such orders would conflict with the First Law.
- A robot must protect its own existence as long as such protection does not conflict with the First or Second Law

Path Planning for Mobile Robots

- Definition: Path planning is the process of deciding a collision-free route from a start point to a goal point for a mobile robot in an environment, possibly cluttered with obstacles.

Classification of Path Planning

- Global Path Planning: Entire map known, planning is “offline” (e.g., Dijkstra’s, A*).
- Local Path Planning: Only nearby environment is known; planning is “online” (e.g., Dynamic Window Approach, Potential Fields).
- Other Classifications:
  - By robot type: holonomic, non-holonomic, kinodynamic
  - By environment: static vs. dynamic; deterministic vs. probabilistic solutions

Types of Obstacles

- Static Obstacles: Stationary, e.g., walls, furniture.
- Dynamic Obstacles: Move unpredictably, e.g., humans, other robots.
- Soft Obstacles: Deformable or can be traversed (e.g., curtains).
- Hard Obstacles: Impassable, solid objects.

Obstacle Avoidance

- Obstacle avoidance enables a mobile robot to reactively modify its trajectory when unexpected obstacles are detected. Techniques range from sensor-based feedback to advanced algorithms like potential fields and DWA.
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The Dynamic Window Approach (DWA) Algorithm

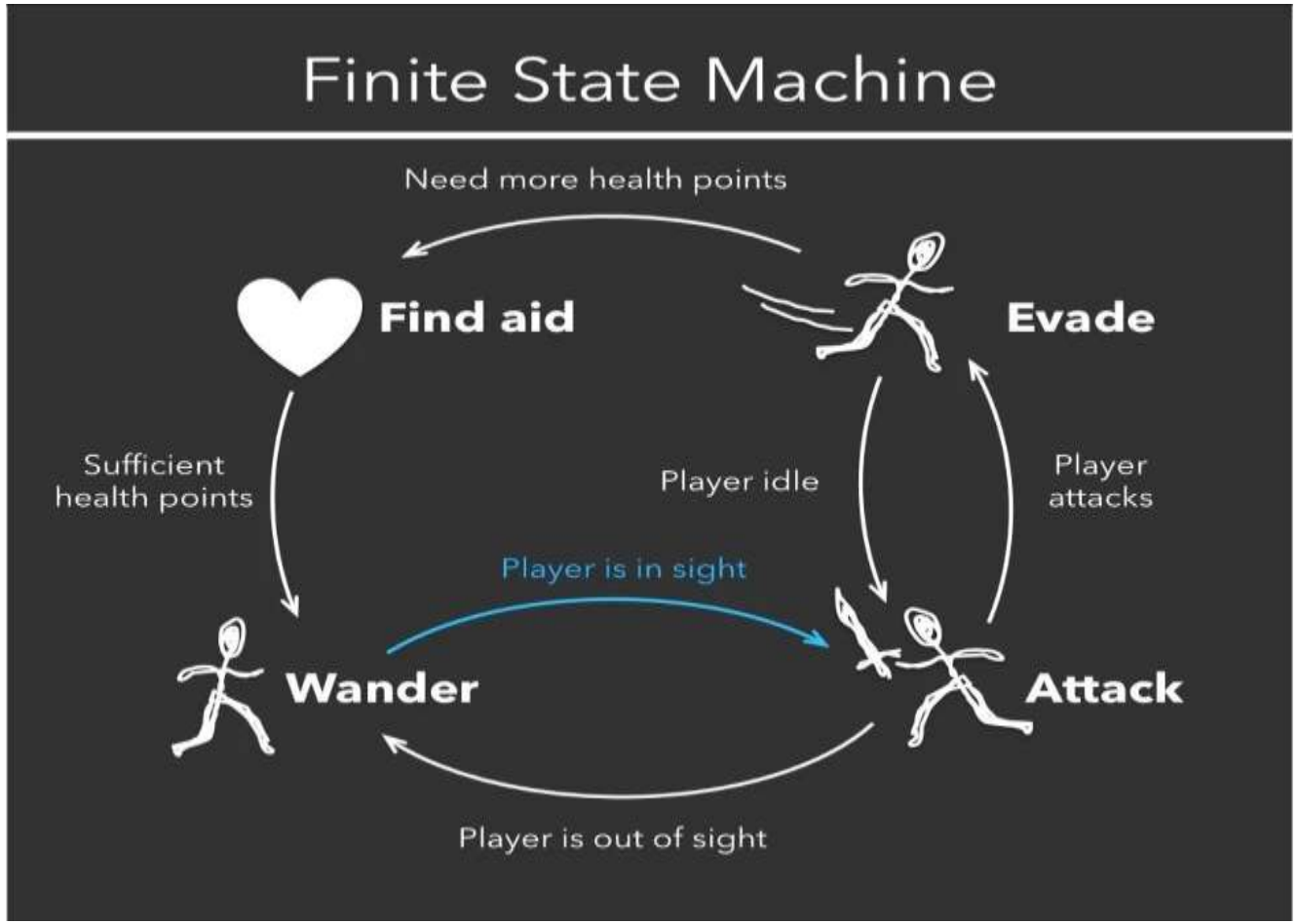
- DWA is a local motion planning algorithm for real-time collision avoidance:
- It samples velocity pairs, evaluates reachable trajectories under constraints, scores based on proximity to target and obstacles, and selects the best control action.
- Widely implemented in the ROS navigation stack as dwa_local_planner.

Visibility Graph for Navigation

- Visibility graph is a graph-based method for optimal pathfinding among polygonal obstacles:
- Nodes represent vertices of obstacles and start/goal positions.
- Edges connect nodes when the direct line is not blocked.
- The shortest path (typically found via Dijkstra's algorithm) is composed of segments between these visible vertices

Artificial Personality

- Emotion State Machine (ESM)
  An **Emotion State Machine** is a computational model that represents emotions as **states** (like happy, sad, angry, neutral), with **transitions** triggered by inputs (stimuli, events, human interaction).


Finite State Machine
Need more health points
Find aid
Evade
Sufficient
health points
Player idle
Player
attacks
Player is in sight
Wander
Attack
Player is out of sight

Creating a Model of Human Behavior

- Creating a model of human behavior involves developing mathematical, computational, or cognitive frameworks that enable robots or AI systems to simulate, anticipate, and react to human actions, emotions, or decisions. Such models are critical for effective human-robot interaction, realistic simulations, personalized assistance, and adaptive collaboration in dynamic environments.

- https://www.youtube.com/watch?v=x7wZCtFWblE
- https://www.youtube.com/watch?v=xuFWuShukaE

Modeling Techniques

1. Cognitive and Psychological Models

- Personality Simulation: Robots can be equipped with simulated personality traits (e.g., the Big Five) and use appraisal theories to generate appropriate emotional responses.
- Decision-Making Frameworks: Approaches like inverse reinforcement learning (IRL) and Bayesian inverse planning are used to infer human goals and reward functions by analyzing observed behavior.

2. Behavioral Observation and Machine Learning

- Imitation Learning & Reinforcement Learning: Robots learn to mimic human actions or optimize their behavior through exploration and feedback.

- Pattern Recognition: Analyzing data from human movement, speech, facial expressions, and gaze for behavioral cues.

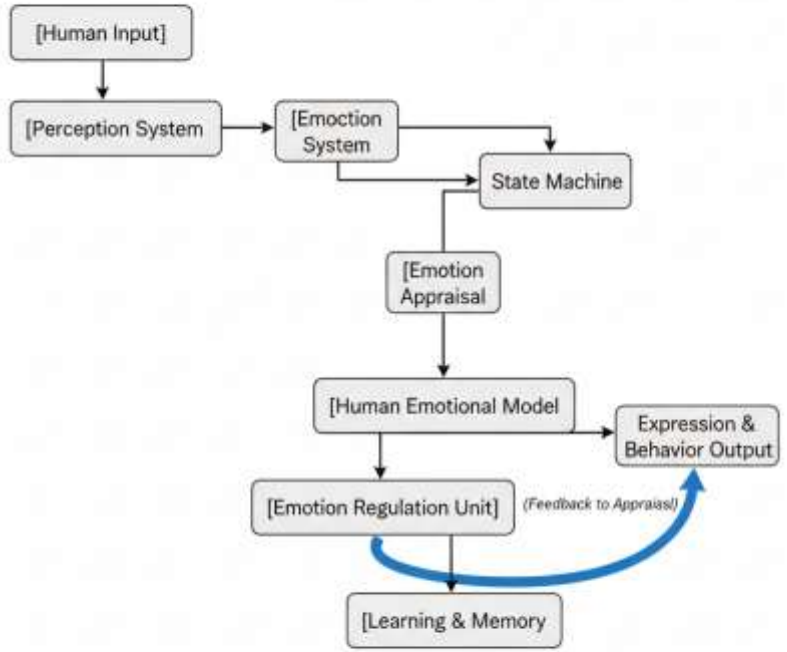
3. Simulation and Digital Twins

- Robotics Simulation: Simulating human behavior virtually allows robots to train or validate their responses in various scenarios, enhancing adaptability and safety without real-world risk.
- Memory and Intent Modeling: Advanced approaches use memory-based learning to build context-aware, adaptive responses and intentional

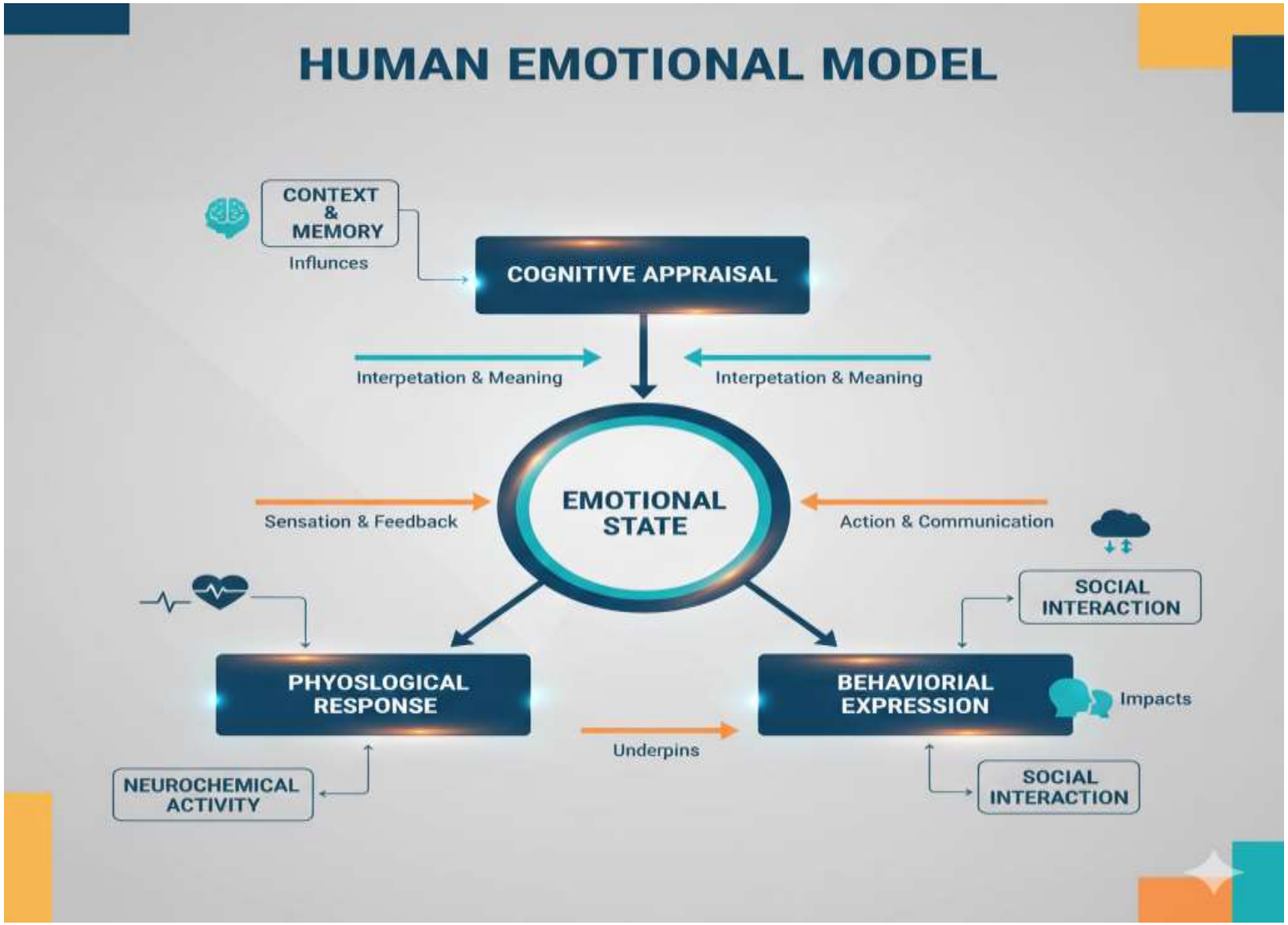
Applications

- Human-Robot Collaboration: Industrial and service robots adapt to worker intentions, gestures, or spoken commands.
- Personalized Assistance: Social robots or virtual agents provide tailored interactions based on user’s emotional state or preferences.
- Training & Simulation: Simulating human users for training autonomous vehicles, customer service bots, and leadership development.
- Healthcare/Assistive Technologies: Robots can predict needs and respond empathetically, e.g., for elderly care or rehabilitation.
- Behavior Analysis/Surveillance: Real-time

Robot Emotion Engine (REE)



- This system processes **Human Input** through a **Perception System** to inform an **Emotion System.**
- The Emotion System and a **State Machine** determine the current emotional status, which is then refined by **Emotion Appraisal.**
- The result feeds into a **Human Emotional Model** to generate the **Expression & Behavior Output** of the robot.
- An **Emotion Regulation Unit** and **Learning & Memory** module help modulate the system, with feedback going back to the Appraisal stage, allowing the robot to learn and adapt its emotional responses over time.


HUMAN EMOTIONAL MODEL
CONTEXT & MEMORY
Influnces
COGNITIVE APPRAISAL
Interpetation & Meaning
Interpetation & Meaning
EMOTIONAL STATE
Sensation & Feedback
Action & Communication
SOCIAL INTERACTION
PHYOSLOGICAL RESPONSE
BEHAVIORIAL EXPRESSION
Impacts
Underpins
NEUROCHEMICAL ACTIVITY
SOCIAL INTERACTION

- The Circumplex Model is a psychological model where emotions are arranged in a circle defined by two primary dimensions:
- **Valence (Pleasure-Displeasure)**: This horizontal axis ranges from negative (unpleasant) to positive (pleasant) feelings.
- **Arousal (Activation-Deactivation)**: This vertical axis ranges from high energy/excitement to low energy/calmness.
- Any emotion, like **anger** (high arousal, negative valence) or **serenity** (low arousal, positive valence), can be plotted as a point within this 2D space. This model effectively shows the **relationship** between different emotional states based on their intensity and pleasantness.